

MITHIL K GOWDA

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Professional Summary

Cybersecurity and AI Security researcher with hands-on experience in secure system design, threat modeling, intelligent cyber defense frameworks, and AI-governed digital ecosystems. Contributor to multiple international research publications, including work in AI-driven cyber defense, intelligent threat detection, carbon credit security systems, and digital trust frameworks. Currently pursuing B.Tech in Information Science with internship experience at ISRO and recognized as a Reliance Foundation Scholar. Interested in Cybersecurity Engineering, Security Research, Threat Intelligence, AI Security, Cloud Security, and Critical Infrastructure Protection.

EDUCATION

M.S. Ramaiah University of Applied Sciences, Bangalore

2022–2026

Bachelor of Technology in Information Science and Engineering

CGPA : 8.8

Technical Skills

Cybersecurity: Network Security, Web Application Security, Threat Modeling, Threat Intelligence, Threat Detection, Vulnerability Assessment, Security Analytics, Security Architecture, Intrusion Detection Systems (IDS).

Cybersecurity Tools: Kali Linux, Wireshark, Nmap, Burp Suite, OWASP ZAP

Programming: Python, Java, JavaScript, SQL, HTML, CSS

Frameworks & Technologies: Flask, Node.js, REST APIs, Git

Databases: MySQL, MongoDB

Research & Analytics: Pandas, NumPy, Data Analytics, Data Visualization, Predictive Modeling

Cloud & Platforms: Google Cloud Platform (GCP)

Operating Systems: Linux, Kali Linux, Windows

Experience

INDIAN SPACE RESEARCH ORGANISATION(ISRO) – Laboratory for Electro Optics Systems (LEOS)

-INTERN *Data Analytics and Full Stack Development*

- Contributing to analytics-driven software development projects involving data processing, visualization, and application design.
- Working on backend development, database integration, and secure application workflows.
- Developing solutions using Python, Flask, and modern web technologies.
- Participating in research-oriented software development activities and data-driven decision-making systems.

Research Publications / Projects

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- Accepted for Oral Presentation at the International Conference on Integrated Intelligence and Cognitive Engineering (ICIICE 2026), Dubai, UAE.
- First Author and Corresponding Author of the book chapter “A HoneyBee-Inspired Metaheuristic Multi-Agent Cyber Defense Framework for Adaptive Deception and Autonomous Response” accepted for publication in the edited volume *Metaheuristic Optimization for Social Good*.
- Proposed and designed a HoneyBee-inspired multi-agent cyber defense architecture integrating intelligent threat detection, adaptive deception, autonomous response, and continuous learning.
- Developed the Detect–Mislead–Neutralize–Learn security lifecycle leveraging XGBoost, CNN-LSTM, Deep Q-Network (DQN) reinforcement learning, adaptive honeypots, and distributed threat intelligence sharing.
- Designed a scalable agent-based cybersecurity framework utilizing Apache Kafka, Flask-based deception environments, machine learning, deep learning, and reinforcement learning for autonomous cyber defense operations.
- Evaluated the framework using CIC-IDS2017 and simulated enterprise attack scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.

- Achieved a deception engagement rate of 92.40% through adaptive attacker redirection and intelligence collection

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract Accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Proposed and designed an AI-governed carbon trading ecosystem integrating Digital Carbon Passports, intelligent governance, fraud detection, trust scoring, and dynamic pricing mechanisms.
- Designed the Secure Trade Authorization and Verification Pipeline (STAVP) leveraging AES-256 encryption, SHA-256 hashing, and secure transaction validation.
- Incorporated machine learning models including Random Forest, XGBoost, and Neural Networks for anomaly detection, trust evaluation, and carbon market intelligence.
- Implemented concepts for secure carbon credit transactions, adaptive pricing, and AI-driven market governance.

Enterprise Canteen Management System

- Developed a centralized enterprise canteen management platform for meal ordering, inventory control, vendor management, and employee attendance tracking.
- Designed and implemented interactive analytics dashboards to support meal demand forecasting, inventory optimization, and procurement cost analysis.
- Integrated Role-Based Access Control (RBAC), secure authentication, and audit logging mechanisms to enhance system security, compliance, and operational transparency.
- Automated manual business processes through real-time reporting, digital record management, and responsive web interfaces, improving efficiency and reducing administrative overhead.

C³T-STAVP Cryptographic Framework (Ongoing Research)

- Developing a hybrid cybersecurity framework integrating Character Chaffing & Transposition Technology (C³T), Secure Transaction Authorization & Verification Protocol (STAVP), and adaptive trust validation mechanisms.
- Researching application-layer semantic obfuscation, ephemeral key generation, Zero-Knowledge Proof (ZKP) verification, and distributed trust architectures for secure digital transactions.
- Investigating privacy-preserving techniques including Crypto-Shredding, off-chain storage, and post-quantum cryptography migration strategies.
- Evaluating framework applications for critical infrastructure protection, financial systems security, and Electronic Health Record (EHR) protection.

Key Research Areas: Cryptography, Cybersecurity, Zero-Knowledge Proofs (ZKP), Trust Architectures, Post-Quantum Security, Privacy-Preserving Systems

Certifications

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| • IBM IT Fundamentals for Cybersecurity Specialization – IBM (Coursera) | 2026 |
| • IBM Front-End Developer Specialization – IBM (Coursera) | 2026 |
| • Introduction to Cyber Security – Indira Gandhi National Open University (SWAYAM) | 2025 |
| • Certified Ethical Hacker (CEH) – Cisco Networking Academy | 2025 |
| • Systems and Usable Security (Elite Certification) – NPTEL | 2025 |
| • Introduction to Internet of Things (Silver Medal) – NPTEL | 2025 |
| • Business Intelligence and Analytics – NPTEL | 2025 |
| • Google Cloud Computing – NPTEL | 2024 |

Achievements/Research Interests

- Reliance Foundation Scholar.
- Research Contributor to International Conference Publications.
- Author/Co-Author of Research Works in Cybersecurity, Artificial Intelligence, and Sustainable Digital Systems.
- Selected for Internship at INDIAN SPACE RESEARCH ORGANISATION(ISRO) – LEOS.